

FORM 2
THE PATENTS ACT 1970
(39 of 1970)
&
THE PATENT RULES, 2003
COMPLETE SPECIFICATION
(See section 10 and rule 13)

1. TITLE OF THE INVENTION: -

DIRT SENSE WASH: ADAPTIVE WASHING LIQUID DETERGENT MECHANISM WITH WEIGHT AND SOIL LEVEL DETECTION MODES

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3. PREAMBLE OF THE DESCRIPTION

The following specification particularly describes the invention and the manner in which it is to be performed

Title of The Invention

Dirt Sense Wash: Adaptive Washing Liquid Detergent Mechanism with Weight and Soil Level Detection Modes.

Field of the Invention

- 5 This invention relates to Dirt Sense Wash: Adaptive Washing Liquid Detergent Mechanism with Weight and Soil Level Detection Modes.

Background of the Invention

Against the backdrop of existing technologies and their limitations, the present invention aims to revolutionize laundry detergent dispensing mechanisms by introducing an adaptive system
10 that optimizes detergent dosage based on both load weight and soil level. By incorporating weight and soil level detection capabilities, the invention offers a more precise and efficient approach to detergent dispensing, ensuring optimal cleaning performance while minimizing detergent waste. This innovative solution enhances user convenience, improves cleaning outcomes, and promotes sustainability by reducing the environmental impact of laundry
15 operations.

In the field of laundry detergent dispensing mechanisms, existing technologies primarily revolve around conventional systems that dispense a fixed amount of detergent regardless of factors such as load size or soil level. These systems often rely on manual measurement or preset settings, leading to inefficiencies and suboptimal cleaning results. Here's an overview of
20 the present state of the art and its associated limitations:

1. Manual Detergent Dispensing:

- **Technology:** Many washing machines and laundry systems require users to manually measure and pour liquid detergent into a dispenser or directly into the washing drum.

- **Limitations:** Manual dispensing can be imprecise and inconsistent, leading to overdosing or under dosing of detergent. This can result in wasted detergent, poor cleaning performance, and potential damage to fabrics. Moreover, users may find it inconvenient and time-consuming to measure detergent for each laundry load, leading to inconsistent cleaning results.

2. Preset Dispensing Settings:

- **Technology:** Some washing machines offer preset dispensing settings that allow users to select a specific amount of detergent based on load size or soil level.

- **Limitations:** While preset settings provide a degree of customization, they may not accurately reflect the actual detergent requirements of each laundry load. Users may still experience overdosing or underdosing of detergent, especially for loads that deviate from standard sizes or soil levels. Additionally, preset settings may lack flexibility and adaptability to changing laundry conditions, leading to suboptimal cleaning outcomes.

3. Sensor-Based Systems:

- **Technology:** Sensor-based detergent dispensing systems use sensors to detect load size or soil level and automatically adjust detergent dosage accordingly.

- **Limitations:** While sensor-based systems offer improved automation and convenience compared to manual or preset dispensing methods, they may still face limitations in accuracy and reliability. Sensors may not always accurately detect load characteristics or soil levels, leading to potential overdosing or underdosing of detergent. Additionally, sensor-based

systems may be more complex and expensive to implement, increasing the cost and maintenance requirements of washing machines.

The invention addresses several key problems inherent in existing laundry detergent dispensing technologies and provides innovative solutions to overcome these limitations:

5 **1. Inefficient Detergent Usage:**

- **Problem:** Conventional detergent dispensing systems often lead to inefficient usage of detergent, with fixed dosages being dispensed regardless of load size or soil level. This can result in wastage of detergent and potential overdosing or underdosing, leading to suboptimal cleaning performance.
- 10 • **Solution:** The invention incorporates weight detection technology to accurately measure the weight of the laundry load. By dispensing detergent proportionally to the load weight, the system ensures that the optimal amount of detergent is used for each load, minimizing wastage and maximizing cleaning efficiency.

2. Inconsistent Cleaning Results:

- 15 • **Problem:** Manual or preset detergent dispensing methods may lead to inconsistent cleaning results, as they do not take into account variations in load size or soil level.
- **Solution:** The invention addresses this issue by offering a soil level detection mode, allowing users to select the appropriate dirtiness level for each load. By adjusting the detergent concentration or washing cycle duration based on the selected soil level, the
20 system ensures thorough cleaning of lightly soiled, moderately soiled, or heavily soiled garments, resulting in consistent and reliable cleaning outcomes.

3. Complexity and Cost of Sensor-Based Systems:

- **Problem:** While sensor-based detergent dispensing systems offer automation and convenience, they may be complex and expensive to implement, leading to higher costs for consumers.

- 5
- **Solution:** The invention combines the simplicity and affordability of weight detection technology with the flexibility and effectiveness of soil level detection, offering a costeffective solution that meets the diverse needs of users. By leveraging these technologies in a user-friendly and intuitive manner, the invention provides an accessible and practical approach to optimizing detergent usage and cleaning performance.

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4. Environmental Impact of Detergent Waste:

- **Problem:** Overdosing or underdosing of detergent not only affects cleaning performance but also contributes to environmental pollution, as excess detergent may end up in wastewater and harm aquatic ecosystems.

- 15
- **Solution:** By accurately dispensing detergent based on load weight and soil level, the invention reduces detergent waste and minimizes environmental impact. This promotes sustainability and eco-friendly laundry practices, aligning with consumer preferences for products that are both effective and environmentally responsible.

None of the prior art indicate above either alone or in combination with one another disclose what the present invention has disclosed. This invention relates to DirtSense Wash: Adaptive

20 Washing Liquid Detergent Mechanism with Weight and Soil Level Detection Modes.

SUMMARY OF THE INVENTION

This summary is provided to introduce a selection of concepts, in a simplified format, that are further described in the detailed description of the invention.

This summary is neither intended to identify key or essential inventive concepts of the invention and nor is it intended for determining the scope of the invention.

To further clarify advantages and features of the present invention, a more particular description of the invention will be rendered by reference to specific embodiments thereof, which is illustrated in the appended drawings. It is appreciated that these drawings depict only typical embodiments of the invention and are therefore not to be considered limiting of its scope. The invention will be described and explained with additional specificity and detail with the accompanying drawings.

Introducing an innovative Adaptive Washing Liquid Detergent Mechanism designed to optimize laundry cleaning efficiency based on two distinct modes: weight detection and soil level detection. In the weight-based mode, the mechanism utilizes a solenoid valve to dispense liquid detergent proportionally according to the weight of the cloth load, ensuring optimal detergent usage and cleaning performance. Additionally, the mechanism features a soil level detection mode, allowing users to select from three dirtiness levels using intuitive buttons: Level 1 for lightly soiled garments, Level 2 for moderately soiled items and Level 3 for heavily soiled clothes. Based on the selected dirtiness level, the mechanism adjusts the detergent concentration or washing cycle duration to effectively remove stains and grime, while minimizing water and detergent consumption. This adaptive approach enhances washing efficiency, reduces detergent waste, and promotes eco-friendly laundry practices, making it an

ideal solution for modern households seeking optimal cleaning results with minimal environmental impact.

DETAILED DESCRIPTION OF THE INVENTION

The detailed description of various exemplary embodiments of the disclosure is described
5 herein with reference to the accompanying drawings. It should be noted that the embodiments
are described herein in such details as to clearly communicate the disclosure. However, the
amount of details provided herein is not intended to limit the anticipated variations of
embodiments; on the contrary, the intention is to cover all modifications, equivalents, and
alternatives falling within the scope of the present disclosure as defined by the appended
10 claims.

It is also to be understood that various arrangements may be devised that, although not
explicitly described or shown herein, embody the principles of the present disclosure.
Moreover, all statements herein reciting principles, aspects, and embodiments of the present
disclosure, as well as specific examples, are intended to encompass equivalents thereof.

15 The terminology used herein is for the purpose of describing particular embodiments only and
is not intended to be limiting of example embodiments. As used herein, the singular forms
“a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly
indicates otherwise. It will be further understood that the terms “comprises,” “comprising,”
“includes” and/or “including,” when used herein, specify the presence of stated features,
20 integers, steps, operations, elements and/or components, but do not preclude the presence or
addition of one or more other features, integers, steps, operations, elements, components
and/or groups thereof.

It should also be noted that in some alternative implementations, the functions/acts noted may occur out of the order noted in the figures. For example, two figures shown in succession may, in fact, be executed concurrently or may sometimes be executed in the reverse order, depending upon the functionality/acts involved.

- 5 In addition, the descriptions of "first", "second", "third", and the like in the present invention are used for the purpose of description only, and are not to be construed as indicating or implying their relative importance or implicitly indicating the number of technical features indicated. Thus, features defining "first" and "second" may include at least one of the features, either explicitly or implicitly.
- 10 Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It will be further understood that terms, e.g., those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or
- 15 overly formal sense unless expressly so defined herein.

The invention of the adaptive washing liquid detergent mechanism with two modes operates on the principle of optimizing detergent usage based on both the weight of the laundry load and the level of soil present on the clothes. Here's a detailed explanation of how the invention works along with working examples:

20 **1. Weight Detection Mode:**

- In this mode, the washing machine or detergent dispenser incorporates a weight detection system that accurately measures the weight of the laundry load before dispensing detergent.

- **Working Example:** When a user initiates a laundry cycle, the washing machine's weight detection system activates and measures the weight of the laundry load placed inside the drum. Based on the weight measurement, the system calculates the appropriate amount of detergent required for optimal cleaning performance.

- For example, if the weight detection system determines that the laundry load is relatively small, it will dispense a proportionally smaller amount of detergent to avoid overdosing. Conversely, for larger loads, a greater amount of detergent will be dispensed to ensure thorough cleaning.

2. Soil Level Detection Mode:

- In this mode, the washing machine or detergent dispenser offers users the option to select the level of soil present on the clothes, ranging from lightly soiled to heavily soiled.

- **Working Example:** The user selects the desired soil level using intuitive buttons or controls on the washing machine or detergent dispenser. For instance, the user may press "Level 1" for lightly soiled clothes, "Level 2" for moderately soiled items, or "Level 3" for heavily soiled garments.

- Based on the selected soil level, the detergent dispenser adjusts the concentration or dosage of detergent to effectively target and remove stains and grime. For lightly soiled clothes, a lower concentration of detergent may be dispensed, while heavily soiled garments may receive a higher concentration for more thorough cleaning.

3. Combination of Modes:

- The invention allows for the combination of both weight detection and soil level detection modes to further optimize detergent usage and cleaning performance.
- **Working Example:** If a user selects "Level 2" soil level for a relatively large laundry load, the detergent dispenser will dispense a moderate amount of detergent based on the weight of the load, adjusted to account for the soil level. This ensures that the detergent dosage is tailored to the specific characteristics of the laundry load, resulting in efficient cleaning without wastage.

4. Adaptive Control and Optimization:

- Throughout the laundry cycle, the detergent dispenser continuously monitors the progress of the wash and may adjust detergent dosage or cycle duration dynamically based on real-time feedback.
- **For example,** if the washing machine detects that the soil level is higher than initially estimated, it may increase the detergent dosage or extend the wash cycle to ensure thorough cleaning. Conversely, if the soil level is lower than expected, it may reduce detergent dosage to prevent wastage

Product Description:

Let's provide an elaborated description of each aspect of the adaptive washing liquid detergent mechanism with two modes:

1. Weight Detection Mode:

- **Technology:** The weight detection mode employs sensors integrated into the washing machine or detergent dispenser to measure the weight of the laundry load before dispensing detergent.

- **Operation:** When a laundry cycle is initiated, the weight detection system activates and measures the weight of the laundry load placed inside the washing machine drum or detergent dispenser compartment.
- 5 • **Algorithm:** Based on the weight measurement, an algorithm calculates the appropriate amount of detergent required for optimal cleaning performance. The algorithm may take into account factors such as load size, fabric type, and detergent concentration to determine the precise dosage.
- 10 • **Dispensing Mechanism:** Once the detergent dosage is calculated, the system triggers the detergent dispensing mechanism, which may utilize a solenoid valve or pump to dispense the liquid detergent into the wash cycle at the appropriate time.

2. Soil Level Detection Mode:

- **Technology:** The soil level detection mode allows users to select the level of soil present on the clothes, ranging from lightly soiled to heavily soiled.
- 15 • **User Interface:** The washing machine or detergent dispenser features intuitive buttons or controls that enable users to indicate the desired soil level for each laundry load.
- **Algorithm:** Upon selecting the soil level, the system adjusts the detergent concentration or dosage based on predefined algorithms tailored to each soil level category.
- 20 • **Adaptation:** The system may dynamically adapt the detergent dosage during the wash cycle based on real-time feedback from sensors monitoring soil level or cleaning progress. For example, if the soil level is higher than initially estimated, the system may increase the detergent dosage to ensure thorough cleaning.

3. Combination of Modes:

- The adaptive washing liquid detergent mechanism allows for the combination of both weight and soil level detection modes to further optimize detergent usage and cleaning performance.
- **Synergy:** By integrating both modes, the system ensures that detergent dosage is tailored to the specific characteristics of each laundry load, such as weight and soil level. This synergistic approach maximizes cleaning efficiency while minimizing detergent waste.
- Example: For a large laundry load with heavy soil, the system may calculate a higher detergent dosage based on weight and adjust the concentration further based on the selected soil level. Conversely, for a small load with light soil, the system may dispense a lower detergent dosage accordingly.

4. User Interface and Controls:

- The washing machine or detergent dispenser features a user-friendly interface with clear controls for selecting detergent dispensing modes and adjusting settings.
- **Display:** A digital display or touchscreen interface provides feedback on selected settings, detergent dosage, and wash cycle progress.
- **Buttons:** Intuitive buttons allow users to navigate through menu options, select soil levels, and initiate laundry cycles with ease.
- **Feedback:** Visual indicators or notifications inform users of the selected detergent dispensing mode, ensuring transparency and user confidence in the cleaning process.

A Dirt Sense Wash: Adaptive Washing Liquid Detergent Mechanism with Weight and Soil Level Detection Modes (100) comprises a weight detection system (101), an algorithm calculation mechanism (102), a detergent dispensing mechanism (103), a soil detection system

(104), an algorithmic adjustment mechanism (105), a mobile application (106), and an alarm system (107), wherein said the weight detection system (101) integrated into a washing machine or detergent dispenser to measure the weight of the laundry load.

In another embodiment the algorithmic calculation mechanism (102) to determine the
5 appropriate detergent dosage based on the measured weight of the laundry load.

In another embodiment the detergent dispensing mechanism activated based on the calculated dosage to dispense liquid detergent into the wash cycle
In another embodiment the soil level detection system (104) allowing users to select the level of soil present on the clothes, ranging from lightly soiled to heavily soiled and dynamic adaptation of detergent dosage during the
10 wash cycle based on real-time feedback from sensors monitoring soil level or cleaning progress.

In another embodiment the algorithmic adjustment mechanism (105) to adapt the detergent dosage based on the selected soil level.

In another embodiment the mobile application (106) works as user interface, where Intuitive
15 controls allowing users to select detergent dispensing modes based on weight detection and soil level detection.

In another embodiment the alarm system (107) and visual indicators used to inform the user to selected detergents dispensing mode and dosage.

We Claim:

1. A Dirt Sense Wash: Adaptive Washing Liquid Detergent Mechanism with Weight and Soil Level Detection Modes (100) comprises a weight detection system (101), an algorithm calculation mechanism (102), a detergent dispensing mechanism (103), a soil detection system (104), an algorithmic adjustment mechanism (105), a mobile application (106), and an alarm system (107), wherein said the weight detection system (101) integrated into a washing machine or detergent dispenser to measure the weight of the laundry load.
2. The laundry detergent dispensing mechanism as claimed in claim 1, wherein the algorithmic calculation mechanism (102) to determine the appropriate detergent dosage based on the measured weight of the laundry load.
3. The laundry detergent dispensing mechanism (103) as claimed in claim 1, wherein the detergent dispensing mechanism activated based on the calculated dosage to dispense liquid detergent into the wash cycle.
4. The laundry detergent dispensing mechanism as claimed in claim 1, wherein the soil level detection system (104) allowing users to select the level of soil present on the clothes, ranging from lightly soiled to heavily soiled and dynamic adaptation of detergent dosage during the wash cycle based on real-time feedback from sensors monitoring soil level or cleaning progress.
5. The laundry detergent dispensing mechanism as claimed in claim 1, wherein the algorithmic adjustment mechanism (105) to adapt the detergent dosage based on the selected soil level.

6. The laundry detergent dispensing mechanism as claimed in claim 1, wherein the mobile application (106) works as user interface, where Intuitive controls allowing users to select detergent dispensing modes based on weight detection and soil level detection.
7. The laundry detergent dispensing mechanism as claimed in claim 1, wherein the alarm system (107) and visual indicators used to inform the users to selected detergents dispensing mode and dosage.

Dated this June 14, 2024



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ABSTRACT

DIRT SENSE WASH: ADAPTIVE WASHING LIQUID DETERGENT MECHANISM WITH WEIGHT AND SOIL LEVEL DETECTION MODES

A Dirt Sense Wash: Adaptive Washing Liquid Detergent Mechanism with Weight and Soil
5 Level Detection Modes (100) comprises a weight detection system (101), an algorithm
calculation mechanism (102), a detergent dispensing mechanism (103), a soil detection system
(104), an algorithmic adjustment mechanism (105), a mobile application (106), and an alarm
system (107), wherein said the weight detection system (101) integrated into a washing
machine or detergent dispenser to measure the weight of the laundry load. In another
10 embodiment the algorithmic calculation mechanism (102) to determine the appropriate
detergent dosage based on the measured weight of the laundry load. In another embodiment
the detergent dispensing mechanism activated based on the calculated dosage to dispense
liquid detergent into the wash cycle. In another embodiment the soil level detection system
(104) allowing users to select the level of soil present on the clothes, ranging from lightly
15 soiled to heavily soiled and dynamic adaptation of detergent dosage during the wash cycle
based on real-time feedback from sensors monitoring soil level or cleaning progress. In
another embodiment the algorithmic adjustment mechanism (105) to adapt the detergent
dosage based on the selected soil level. In another embodiment the mobile application (106)
works as user interface, where Intuitive controls allowing users to select detergent dispensing
20 modes based on weight detection and soil level detection. In another embodiment the alarm
system (107) and visual indicators used to inform the user to selected detergents dispensing
mode and dosage.